AYUSH SONI

Senior Data Scientist — Causal Inference, Experimentation, ML & GenAI/LLM Systems

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SUMMARY

Senior Data Scientist with 6+ years building causal inference, experimentation, and ML systems at enterprise scale. Designs and runs quasi-experimental causal analyses to isolate true promotional lift, builds production forecasting models (SARIMAX, XGBoost), and ships GenAI agentic systems on Snowflake Cortex AI and Databricks-hosted LLMs — including a promotional effectiveness agent with statistical tool-calling, a Text2SQL natural-language query layer, and a causal demand-planning system. Built and validated a transformer-based NLP classifier (FinBERT, 87% accuracy) at UBS. Built and evaluated a financial document intelligence agent with reranked RAG, Text2SQL, Claude tool calling, FastAPI, Docker, and a 15-question evaluation harness. Seeking Senior Data Scientist, Applied Scientist, ML Engineer, and AI Engineer roles spanning experimentation, ML, and GenAI systems.

CORE TECHNICAL SKILLS

ML & Causal Modeling: Causal Inference & Demand Forecasting, Quasi-Experimental Design, Time-Series (SARIMAX), XGBoost, Regression & Classification, Random Forest, Deep Learning (PyTorch, TensorFlow, Keras), Recommender Systems

Experimentation & Analytics: A/B Testing, Promotional Lift & Causal Decomposition, Seasonality/Trend/DOW/Holiday Decomposition, Statistical Inference (t-tests, ANOVA, regression), Large-Scale Data Analysis

GenAI & LLM Engineering: GenAI Agentic Workflows, Text2SQL, LLM Tool-Calling (Snowflake Cortex AI, Databricks-hosted LLMs), Stored-Procedure Orchestration, NLP/Transformers (FinBERT, Hugging Face)

LLM Evaluation: LLM-as-judge, Faithfulness/Relevance Scoring, Automated Evaluation Frameworks, Prompt-based Evaluation

MLOps & Cloud: MLflow, Docker, Kubernetes, Airflow, Jenkins, CI/CD, AWS (S3, EC2, SageMaker, Redshift), GCP (BigQuery, Dataflow), Azure (Data Factory, SQL Database)

Data Platforms & Engineering: Snowflake, Databricks, PySpark, Hive, ETL/ELT Pipeline Architecture

Programming & Databases: Python, SQL, R, Base SAS, Git/GitHub

PROFESSIONAL EXPERIENCE

DISH / EchoStar (Boost Mobile) — Denver, CO

Senior Data Scientist — Sep 2024 – Present

Leads AI/ML lifecycle for the Retail Wireless Product Data & Analytics team, building causal inference models, GenAI agentic pipelines, and experimentation-driven analytics across Indirect, Web, and Walmart sales channels.

- Built a **Snowflake Cortex AI promotional effectiveness agent** — accepts natural-language inputs (device, promo type, dates), orchestrates a **Python stored procedure** that decomposes and removes confounders (seasonality, day-of-week, trend, holiday, new-device halo) from sales data, and returns structured **pre/promo/post lift tables** with LLM-generated narrative takeaways — a full agentic tool-calling pipeline on Snowflake Cortex.
- Designed and executed **4-week quasi-experimental causal analysis** of iPhone Flash Pricing — normalizing for weekday mix, SSI, trend, and seasonality to isolate true incremental lift, quantifying **+27% GA lift** for iPhone 16e and **+1,083% lift** for iPhone 17e, and identifying a structural web conversion suppressor through pre/promo/post decomposition.
- Built an integrated **causal AI demand-planning system** (SARIMAX + XGBoost) across three sales channels, exceeding a **75% N-4 forecast-accuracy** target through channel-specific modeling and causal feature engineering on pricing and promotion signals.
- Deployed a **Text2SQL natural-language query layer** using **Snowflake Cortex AI**, enabling business users to query a multi-domain product data ecosystem (activations, pricing, inventory, subscriber lifecycle) in plain English — full SQL-to-English lifecycle from schema grounding to answer delivery.
- Built and deployed **GenAI agents** on **Databricks-hosted LLMs** and Snowflake Cortex AI enabling natural-language KPI queries and automated executive reporting — reducing analytics turnaround time by **70%**.
- Architected a **4-phase Snowflake MLOps deployment framework** governing dataset onboarding, staging validation, YAML-based production promotion, and post-deployment monitoring and feedback loops.
- Engineered executive-facing analytics platform (Tableau + Snowflake, LOD calculations, in-stock prediction) that accelerated decision-making speed by **30%** and cut manual reporting effort by **40%**.
- Partnered cross-functionally with marketing, supply chain, and retail ops — achieving **100% accuracy** in executive KPI reporting.

The Partnership Center Ltd. — Cincinnati, OH

Data Scientist — Jun 2024 – Sep 2024

- Deployed a **Python-based regression model** into production to optimize resource utilization, reducing forecast error by **20%** and improving schedule-tracking efficiency by **40%**.
- Migrated enterprise data to Snowflake and automated **ETL workflows** (Alteryx + Snowflake) into repeatable, production-ready pipelines for portfolio performance monitoring.
- Built interactive **Power BI dashboards** with advanced DAX calculations surfacing investment-portfolio performance trends for portfolio managers.
- Partnered cross-functionally to operationalize ML outputs into core business processes for resource planning.

UBS — Chicago, IL

Data Scientist — Quantitative Research & NLP/AI — Jan 2023 – May 2024

Advanced quantitative research and risk modelling by applying AI/ML forecasting, NLP (FinBERT), and automation to financial datasets.

- Revamped **XGBoost** regression models on **24 months'** rolling data, achieving a **62.40% hit rate** on weekly bond-market predictions and strengthening investment decision support.

- Built and deployed an **NLP sentiment-classification model** using **FinBERT** (transformer-based LLM), achieving **87% accuracy** classifying financial news and feeding outputs directly into credit-risk assessments.
- Automated portfolio data ingestion from **YODA and Bloomberg** via Python scripting, cutting manual effort by **70%** and improving governance model-validation efficiency for US multi-sector teams.
- Improved global mutual-fund forecasting accuracy by **33%** through advanced data validation and feature engineering on Bloomberg datasets.
- Designed predictive pipelines (Python: NumPy, Pandas, Scikit-Learn, TensorFlow, Keras; SQL) to model equity performance and inform quantitative trading strategies.

Accenture Pune, India
Machine Learning Associate Jun 2018 – Jul 2021

Client: OP Financial, Finland

- Built dynamic credit/debit card limit-management models (account activity, loan history, credit scores), increasing revenue by **23%** and operational efficiency by **40%**.
- Applied **K-means clustering** and rule-based segmentation by risk-adjusted revenue potential; conducted **A/B tests** on ML-driven customer segments to improve campaign satisfaction and response rates.
- Applied statistical inference (**t-tests, chi-square, ANOVA, regression**) to uncover patterns in credit/debit portfolios.
- Built Tableau dashboards and SQL views tracking financial KPIs, improving market-data accuracy by **25%**.

Client: Allianz Bank, Germany

- Developed an **XGBoost classifier** for customer churn prediction (**F1: 0.84**), identifying key loyalty drivers and informing retention strategy.
- Built a Python regression model improving resource-utilization forecasts (20% error reduction, 40% scheduling-efficiency gain).
- Migrated enterprise data from **DB2/Teradata to AWS Cloud**, ensuring scalable, cost-effective, and reliable analytics infrastructure.
- Standardized experimentation frameworks to minimize bias and improve model interpretability, increasing analytical accuracy by **90%**.

PROJECTS

Financial Document Intelligence Agent (In Progress)

- Built a reranked RAG pipeline over SEC filings using FAISS, sentence-transformer embeddings, and a cross-encoder, with ticker-aware retrieval across five financial institutions.
- Built a Claude tool-calling router that separates filing search from read-only Text2SQL over XBRL facts and price history, with FastAPI, Docker, and a Streamlit interface verified locally.
- Built a 15-question evaluation harness with deterministic grounding checks and LLM-as-judge scoring; recorded 5/5 faithfulness for SQL-backed numeric answers and a 2-4/5 narrative limitation.

Beijing Air Quality Index Forecasting

- Formulated a machine learning-based air-quality prediction strategy generating a **24-hour forecast** — implemented Neural Network, Gaussian Naïve Bayes, Linear Regression (Gradient Descent), and Multi-class Logistic Regression, evaluated by RMSE after hyperparameter tuning.
- **Selected as one of the top final projects** and hosted on the course website for exemplary work, recognised by Professor.

NLP Sentiment Analysis for Financial News (FinBERT Prototype)

- Implemented **FinBERT** (transformer-based LLM) to classify financial-news sentiment with **87% accuracy**; integrated outputs into credit-risk assessments.

CERTIFICATIONS & CONTINUING EDUCATION

- Meta Data Analyst, issued by Meta
- Google Advanced Data Analytics, issued by Google
- Data Visualization in Tableau & Power BI, Udemy
- Continuing study in production LLM evaluation, observability, adversarial testing, and retrieval quality
- Q2 FY20 Accenture Team Award (Client & Customer category) — reduced technical issues by 84%

EDUCATION

Northeastern University Boston, MA
M.S. Data Analytics Engineering Sep 2021 – Dec 2023
Coursework: Data Management, Data Visualization, Statistical Methods, Data Mining, Machine Learning, Deep Learning

Narsee Monjee Institute of Management Studies Mumbai, India
B.Tech in Computer Engineering